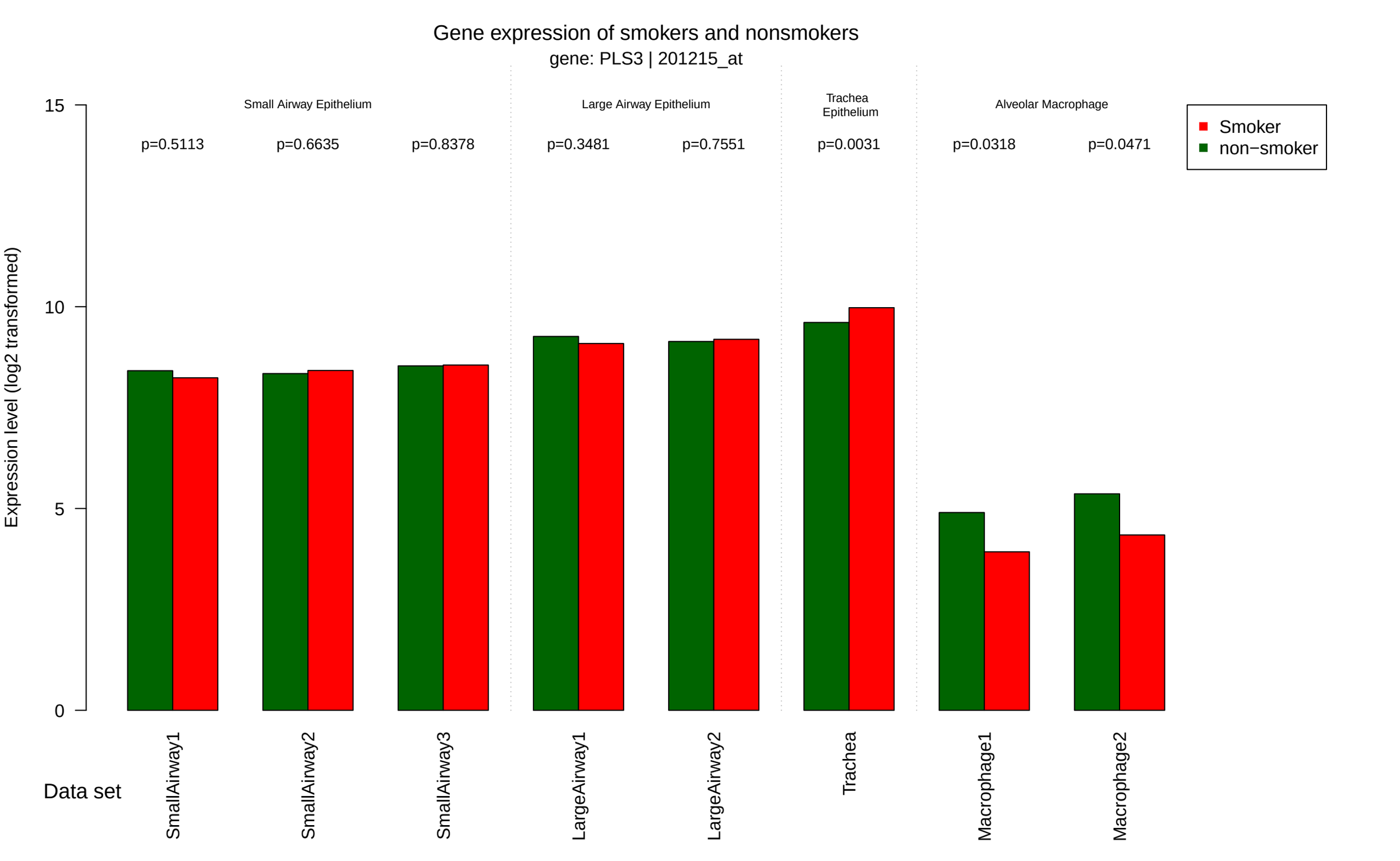
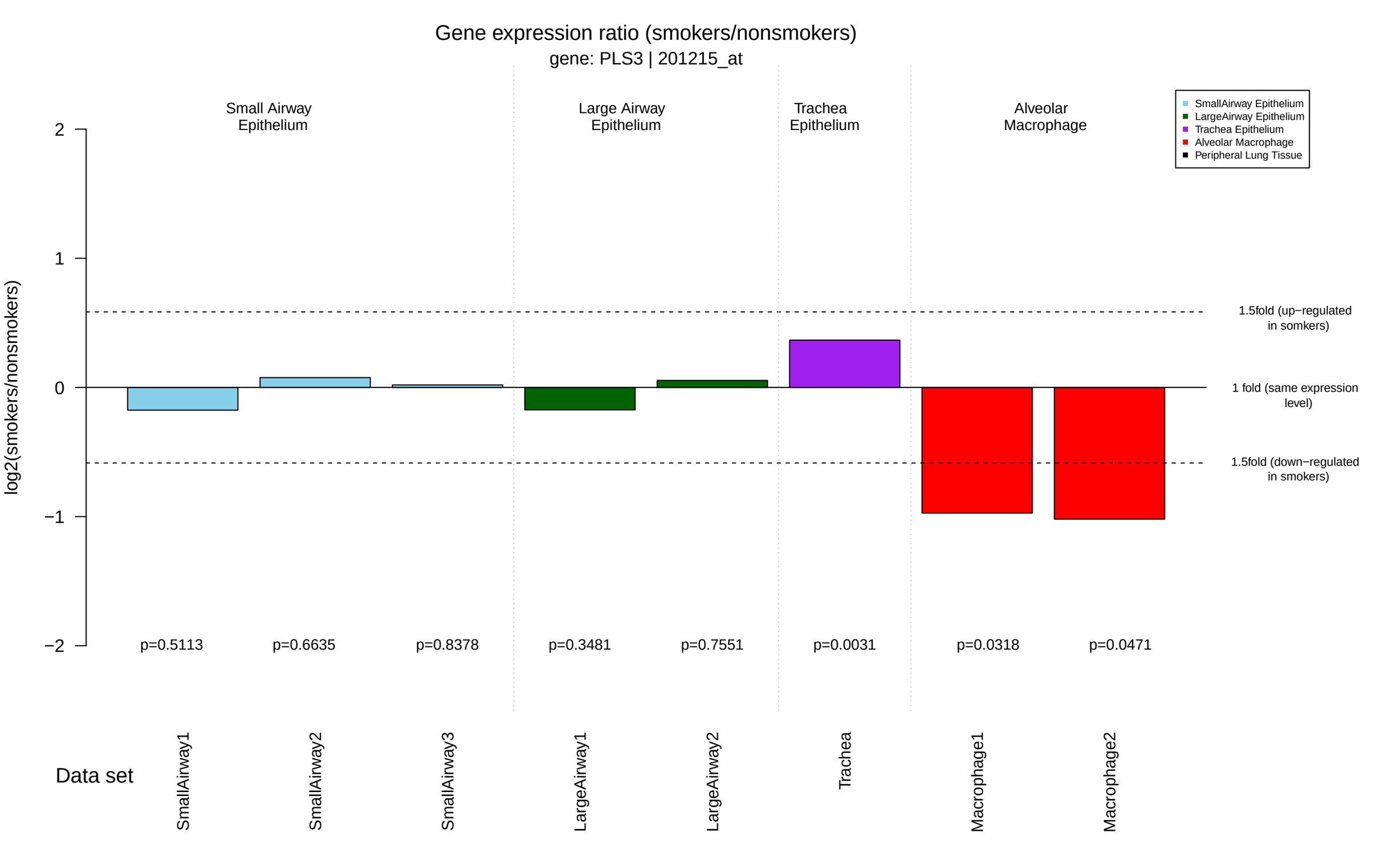
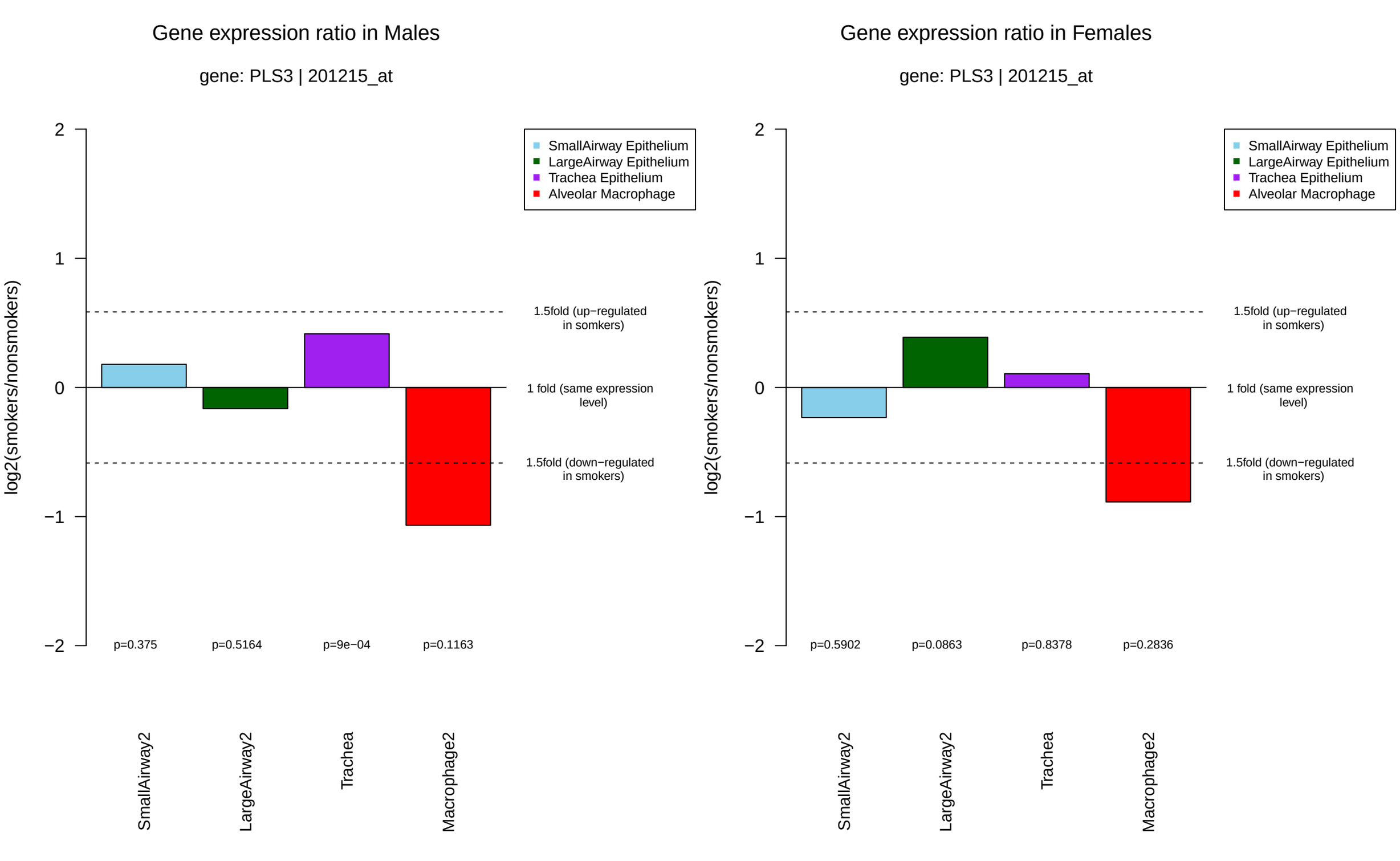
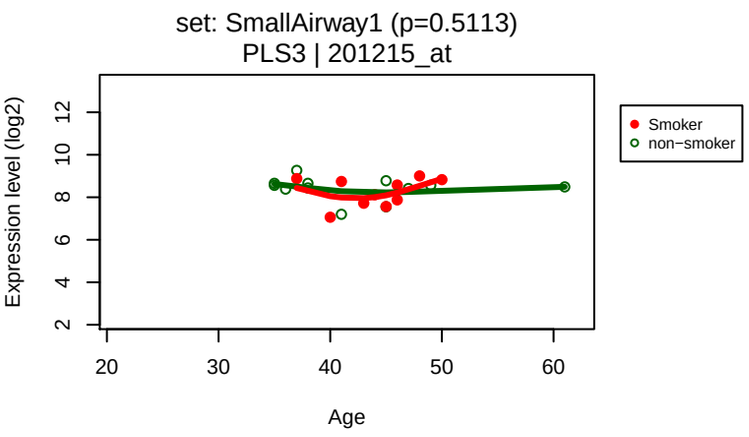
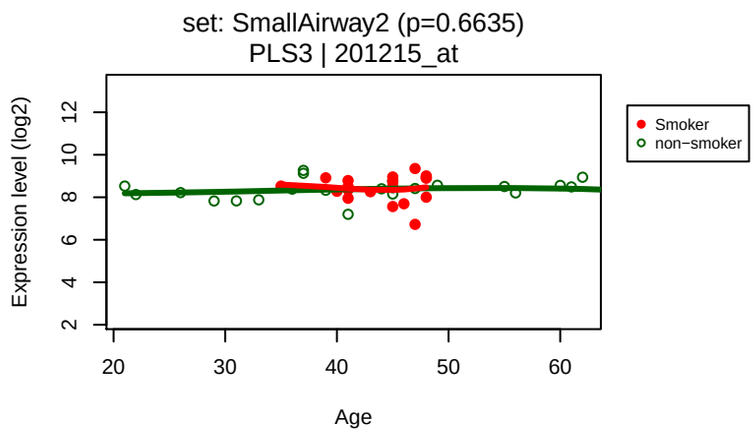
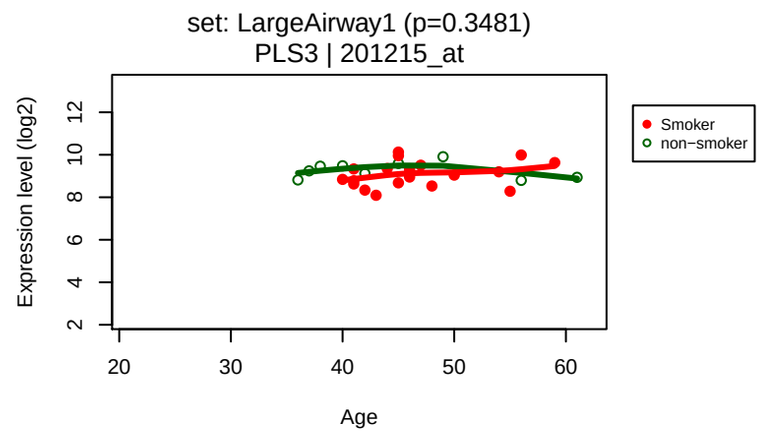

Gene expression of smokers and nonsmokers
gene: PLS3 | 201215_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.5113
p=0.6635
p=0.8378
p=0.3481
p=0.7551
p=0.0031
p=0.0318
p=0.0471
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

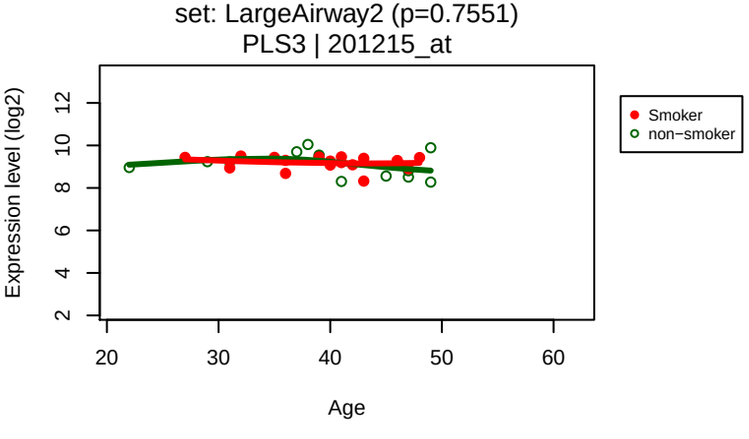

Gene expression ratio (smokers/nonsmokers)
gene: PLS3 | 201215_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5113
p=0.6635
p=0.8378
p=0.3481
p=0.7551
p=0.0031
p=0.0318
p=0.0471
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

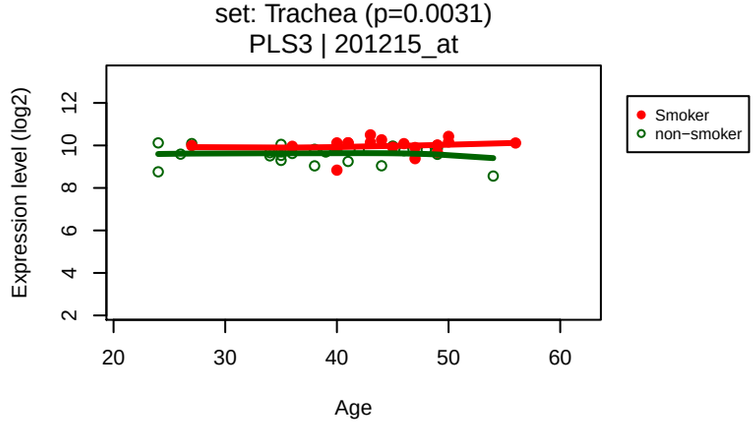

Gene expression ratio in Males
gene: PLS3 | 201215_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.375
p=0.5164
p=9e−04
p=0.1163
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PLS3 | 201215_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5902
p=0.0863
p=0.8378
p=0.2836
SmallAirway2
LargeAirway2
Trachea
Macrophage2

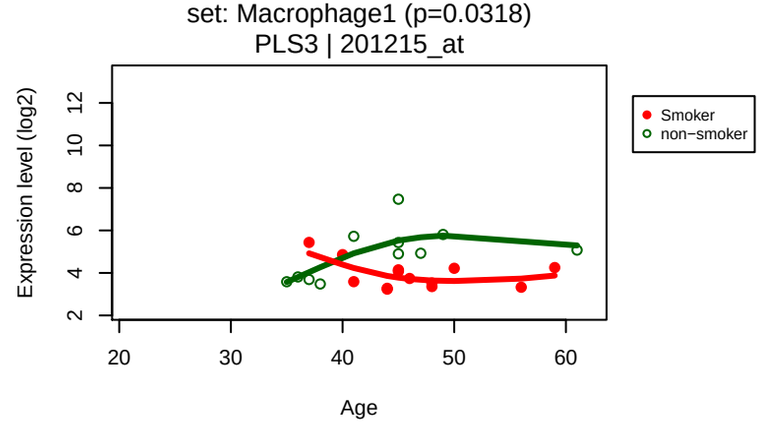

set: SmallAirway1 (p=0.5113)
PLS3 | 201215_at
Expression level (log2)
Age
Smoker
non−smoker

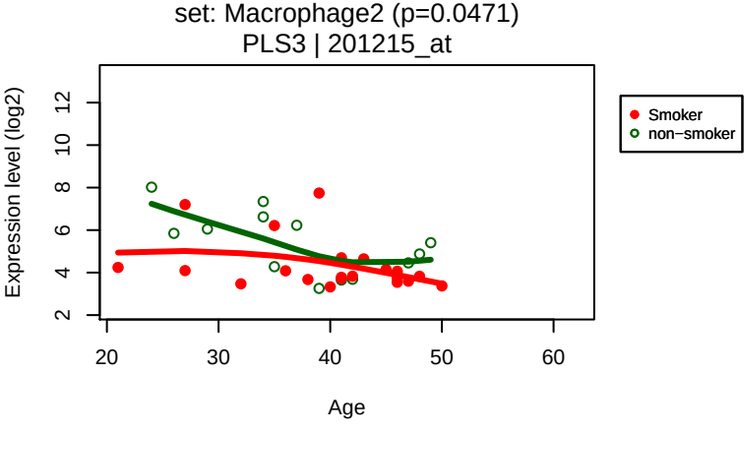

set: SmallAirway2 (p=0.6635)
PLS3 | 201215_at
Expression level (log2)
Age
Smoker
non−smoker

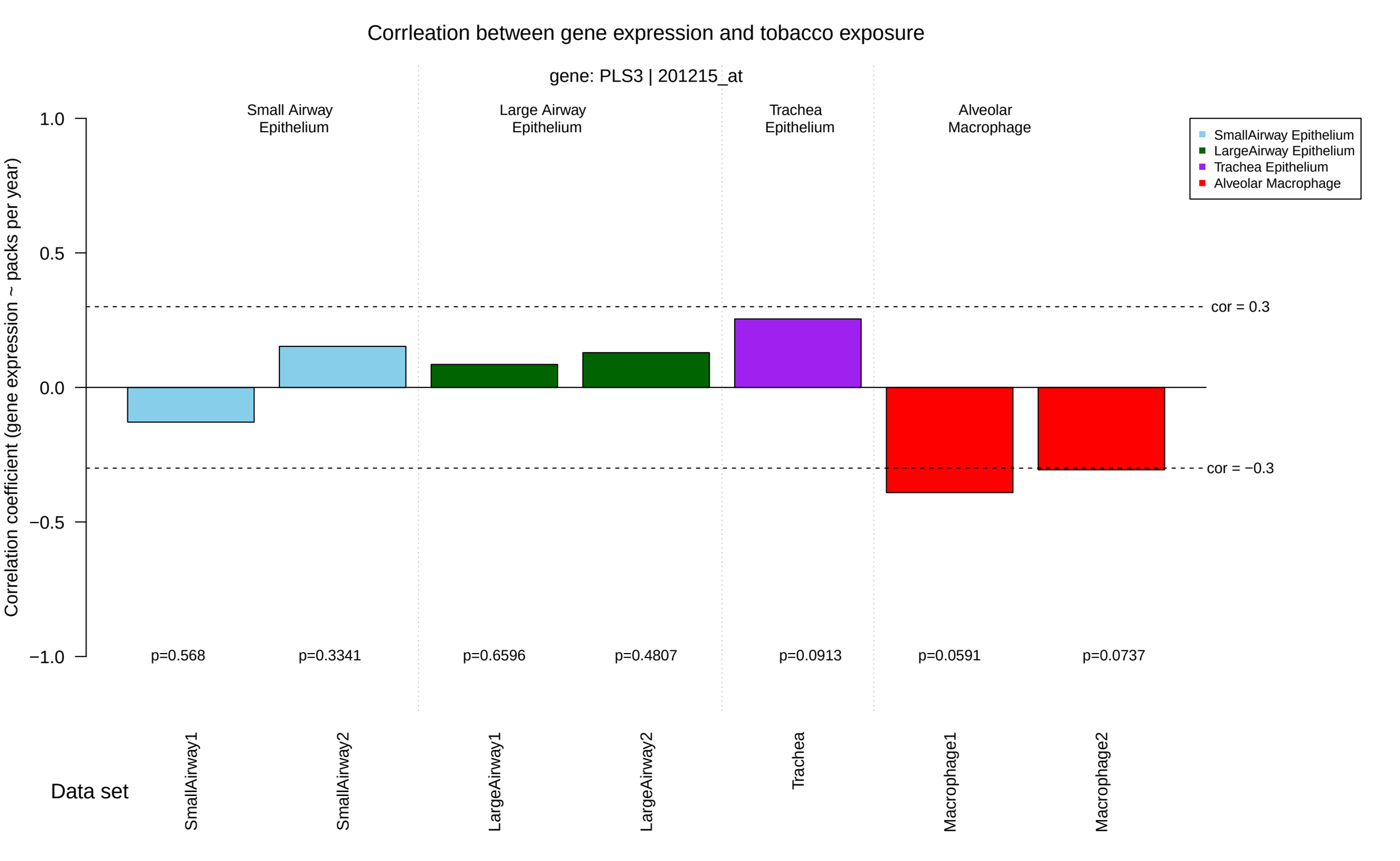

set: LargeAirway1 (p=0.3481)
PLS3 | 201215_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.7551)
PLS3 | 201215_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0031)
PLS3 | 201215_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0318)
PLS3 | 201215_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0471)
PLS3 | 201215_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PLS3 | 201215_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.568
p=0.3341
p=0.6596
p=0.4807
p=0.0913
p=0.0591
p=0.0737
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2